

# MAHANTESH KHETRI

Laser Process Development | Semiconductor Materials | Photonics | Automation

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## PROFESSIONAL PROFILE

PhD candidate in Electrical and Computer Engineering at the University of Virginia, expected December 2026. Develops laser processes using picosecond, nanosecond, and continuous-wave systems, with experience in semiconductor and photovoltaic processing, thin films, materials characterization, thermal modeling, and Python-based process automation. Four first-author laser process papers and co-inventor on a published US patent application.

## TECHNICAL EXPERTISE

**Laser systems and processes:** UV, visible, and IR lasers; selective ablation, debonding, delamination, surface modification; optical alignment, galvo scanning, F-theta optics, EZCAD, and process-window development.

**Materials and metrology:** PLD, thermal evaporation, high-vacuum systems, single-crystal growth; SEM/EDS, TEM, XRD, XPS, Raman, FTIR, UV-Vis, AFM, PPMS, carrier lifetime, and electrical measurements.

**Modeling and automation:** Python, NumPy, pandas, SciPy, Matplotlib, OpenCV, transient thermal modeling, U-Net/CNN segmentation, image-to-vector conversion, and laser scan-path generation.

## RESEARCH AND ENGINEERING EXPERIENCE

### University of Virginia | Doctoral Researcher / Graduate Research Assistant

August 2022 - Present | Charlottesville, Virginia

- Develop and optimize selective laser processing for silicon, CIGS, metallic contacts, and multilayer photovoltaic structures; connect processing conditions with throughput and substrate integrity.
- Build, align, calibrate, and troubleshoot galvo-scanned laser systems and beam-delivery hardware. Characterize outcomes through microscopy, spectroscopy, and electrical measurements.
- Develop Python and OpenCV workflows for thermal analysis, image segmentation, geometry extraction, and automated vector scan-path generation.
- Document experiments, prepare technical publications and presentations, and train laboratory users on equipment operation.

### Indian Institute of Technology Delhi | Research Fellow

December 2019 - July 2022 | New Delhi, India

- Performed thin-film and materials research using pulsed laser deposition, single-crystal growth, and high-vacuum equipment. Used structural, surface, and transport measurements to connect growth conditions with material properties.

### Indian Institute of Technology Ropar | Research Student

July 2018 - May 2019 | Rupnagar, India

- Fabricated Sb-doped SnO<sub>2</sub> films by thermal evaporation and controlled post-oxidation; evaluated structure, optical transmission, and electrical resistivity.

# MAHANTESH KHETRI

Education, selected research outputs, and recognition

## EDUCATION

**PhD, Electrical and Computer Engineering** | University of Virginia

August 2022 - Expected December 2026 | Advisor: Prof. Mool C. Gupta

**MSc, Physics** | Indian Institute of Technology Ropar | 2017 - 2019

**BSc, Computer Science, Physics, and Mathematics** | Rani Channamma University | 2013 - 2016

University Second Rank; Gold Medal in Computer Science

## SELECTED PUBLICATIONS

**Laser removal of silicon solar cell backsheet while preserving tempered glass and silicon wafer**

Mahantesh Khatri, Pawan K. Kanaujia, Mool C. Gupta. *Solar Energy Materials and Solar Cells* (2026).

doi: 10.1016/j.solmat.2026.114404

**Laser recycling of silver in bulk and nanoparticle form from silicon solar cells and deep learning for process automation**

Mahantesh Khatri, Pawan K. Kanaujia, Mool C. Gupta. *Renewable Energy* (2025).

doi: 10.1016/j.renene.2025.123684

**Recycling of silver from CIGS solar cells using laser ablation and debonding**

Mahantesh Khatri, Mool C. Gupta. *Solar Energy Materials and Solar Cells* (2025).

doi: 10.1016/j.solmat.2024.113381

**Recycling of silver from silicon solar cells by laser debonding**

Mahantesh Khatri, Mool C. Gupta. *Solar Energy* (2024).

doi: 10.1016/j.solener.2024.112381

**Stabilization of 5 wt percent Sb doped SnO<sub>2</sub> thin film by post oxidation of thermally evaporated metallic layer**

Mahantesh Khatri, R. Ramarajan, N. Purusotham Reddy, M. Kovendhan, K. V. A. Srivatsav, B. S. Rao, K. Thangaraju, D. Paul Joseph. *AIP Conference Proceedings* (2019).

doi: 10.1063/1.5113117

## PUBLISHED PATENT APPLICATION

**Use of Lasers to Selectively Remove Materials from Substrates**

Mool C. Gupta and Mahantesh Khatri | Co-inventor

US20260008129A1 | Published January 8, 2026

## SELECTED RECOGNITION

- Wilson Bicentennial Graduate Fellowship, University of Virginia, 2025-2026.
- First Place, Electrical Engineering Poster Presentation, UVERS 2025.
- UPLAMP 2025 Travel Award, Vilnius, Lithuania.
- NSF-supported ANGEL-7 Travel Award, 2024.